

Power Converter Analysis and Design

ASSIGNMENT - 4

1. In a three-phase three-wire system excited with voltage sources,
 - (a) positive sequence currents are absent, while negative and zero sequence currents may be present.
 - (b) negative sequence currents are absent, while positive and zero sequence currents may be present.
 - (c) zero sequence currents are absent, while positive and negative sequence currents may be present.
 - (d) currents of all three sequences may be present.
2. Consider an ideal 3-leg (6 switches) 2-level 3-phase inverter with the PWM modulation as shown in (1), with a variable modulation index m

$$\begin{aligned}d_1 = 1 - d_4 &= \frac{1}{2} + \frac{m}{2} \sin(\omega t), \\d_3 = 1 - d_6 &= \frac{1}{2} + \frac{m}{2} \sin\left(\omega t - \frac{2\pi}{3}\right) \\d_5 = 1 - d_2 &= \frac{1}{2} + \frac{m}{2} \sin\left(\omega t + \frac{2\pi}{3}\right).\end{aligned}\tag{1}$$

Modulation using the above duty ratio scheme results in distortion in the leg output voltage patterns for $m > 1$. However, a slight modification in modulation index, as shown in (2), results in the output voltage (line-to-line) free of distortion.

$$\begin{aligned}d_1 = 1 - d_4 &= \frac{1}{2} + \frac{m}{2} \sin(\omega t) + \frac{m}{12} \sin(3\omega t), \\d_3 = 1 - d_6 &= \frac{1}{2} + \frac{m}{2} \sin\left(\omega t - \frac{2\pi}{3}\right) + \frac{m}{12} \sin(3\omega t), \\d_5 = 1 - d_2 &= \frac{1}{2} + \frac{m}{2} \sin\left(\omega t + \frac{2\pi}{3}\right) + \frac{m}{12} \sin(3\omega t).\end{aligned}\tag{2}$$

- (a) Show that the revised duty ratio selection in (2) yields the same line-to-line output voltages as in (1) while in linear range of operation.
- (b) Find an allowable limit on the modulation index m to avoid saturating duty ratios in (2).

3. In a selective harmonic elimination approach for two-level inverters, what would the minimum number of switchings needed per quarter-cycle in the voltage waveform to eliminate the harmonics of orders 5, 7, 11, 13, 17, and 19?
4. An ideal three-phase three-phase voltage source inverter (VSI) is operated with the switching waveform shown in Fig. 1, designed with switching angles $\alpha_1 = 10.55^\circ$, $\alpha_2 = 16.09^\circ$, $\alpha_3 = 30.91^\circ$ and $\alpha_4 = 32.87^\circ$. Determine the magnitudes of the fundamental, 3rd, 5th, 7th, 11th, and 13th harmonics in the line-to-line voltages v_{ab} , v_{bc} and v_{ca} .

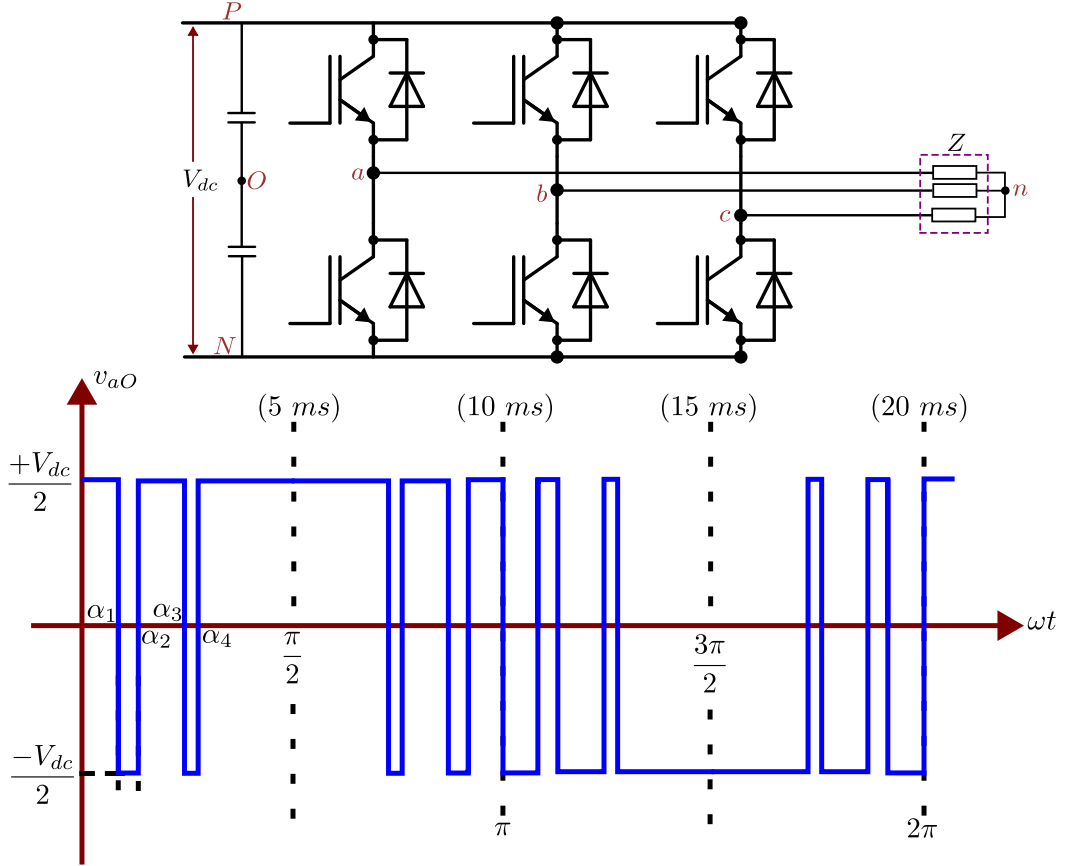


Figure 1: 3-Phase voltage source inverter (VSI) with Selective Harmonic Elimination (S.H.E) scheme

5. The performance difference(s) between the unipolar and bipolar switching schemes applied to a single-phase H-bridge inverter fed from a given DC bus is(are):
 - (a) The maximum amplitude of the fundamental sine wave in v_{ab} (where a and b are the switching nodes of two legs) is greater in the bipolar scheme than in the unipolar scheme.
 - (b) The voltage contains a component at the switching frequency in the bipolar scheme, while this is not the case in the unipolar scheme.
 - (c) The current ripple for a given R-L load is higher in the bipolar scheme than in the unipolar scheme for the same fundamental output voltage.
 - (d) The device switching transition time in the case of bipolar modulation is larger than in the case of unipolar modulation.

6. An ideal three-phase voltage source inverter (VSI) is shown in Fig. 2. It is required to generate a space vector with the following characteristics:

- (a) Amplitude 0.5 at an angle of 15° (anticlockwise) from the a-axis.
- (b) Amplitude 0.5 at an angle of 150° (anticlockwise) from the a-axis.

For each case, determine:

- (i) What are the bounding states corresponding to each space vector?
- (ii) The duration for which the inverter should remain in the bounding states to produce the equivalent space vector over a period of $50 \mu\text{s}$.

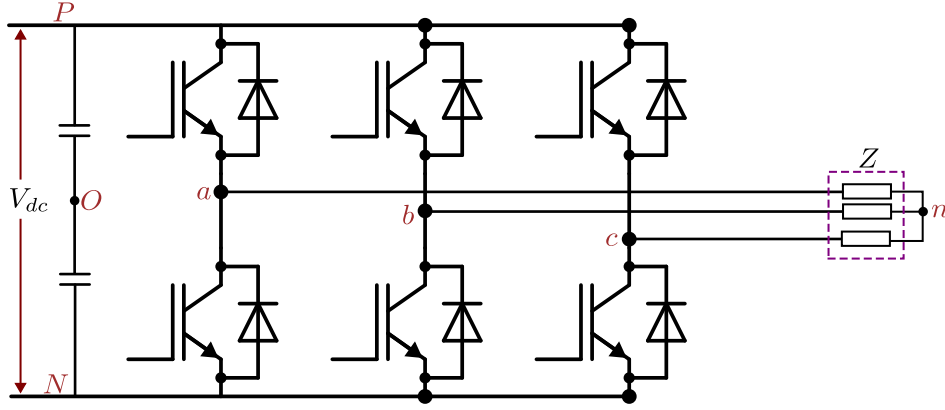


Figure 2: 3-Phase voltage source inverter (VSI)

7. An ideal three-phase voltage source inverter (VSI) is shown in Fig. 2. It is required to generate a space vector at an angle of 30° (anti-clockwise) from the a-axis such that the vector tip just touches the hexagonal boundary. How much time should the inverter system spend in the bounding states to generate this equivalent space vector in a period of $50 \mu\text{s}$? What are the bounding states?
8. The three-phase voltages in a power supply system are given by

$$v_a = 325 \cos(\omega t - 22.5^\circ) \text{ V}, \quad v_b = 325 \cos(\omega t - 142.5^\circ) \text{ V}, \quad v_c = 325 \cos(\omega t - 262.5^\circ) \text{ V}.$$

Find the space vector of voltages at a time instant $t = 12 \text{ ms}$ in terms of magnitude and angle, given that $\omega = 100\pi \text{ rad/s}$. Additionally, if this must be realized using a two-level inverter, determine which space vectors will be used.

9. Consider a three-phase system with currents in the abc frame as shown below. Convert these currents to the $\alpha\beta 0$ frame using Clarke transformation.

- (a)

$$i_a(t) = 5 \cos(\omega t - \phi), \quad i_b(t) = 5 \cos\left(\omega t - \phi + \frac{2\pi}{3}\right), \quad i_c(t) = 5 \cos\left(\omega t - \phi + \frac{4\pi}{3}\right).$$

- (b)

$$i_a(t) = 5 \cos(\omega t - \phi), \quad i_b(t) = 4.9 \cos\left(\omega t - \phi + \frac{2\pi}{3}\right), \quad i_c(t) = 5 \cos\left(\omega t - \phi + \frac{4\pi}{3}\right).$$

10. Consider an ideal three-phase voltage source inverter (VSI) shown in Fig. 2. The switching node to DC bus midpoint voltages (v_{aO} , v_{bO} , v_{cO}) and the switching node to DC bus negative voltages (v_{aN} , v_{bN} , v_{cN}) are shown in Fig. 3.

- (i) The VSI is switched using a unipolar SPWM scheme.
- (ii) The VSI is switched using a bipolar SPWM scheme.
- (iii) In the frequency spectrum of voltage v_{ab} a harmonic component will appear at $7f_{carrier} \pm f_{modulation}$
- (iv) In the frequency spectrum of voltage v_{ab} a harmonic component will appear at $f_{carrier} \pm 2f_{modulation}$

Identify the correct statement regarding the VSI switching:

- (a) Statements (i), (iii), and (iv) are true
- (b) Statements (ii), (iii), and (iv) are true
- (c) Statements (i) and (iv) are true
- (d) Statements (ii) and (iii) are true

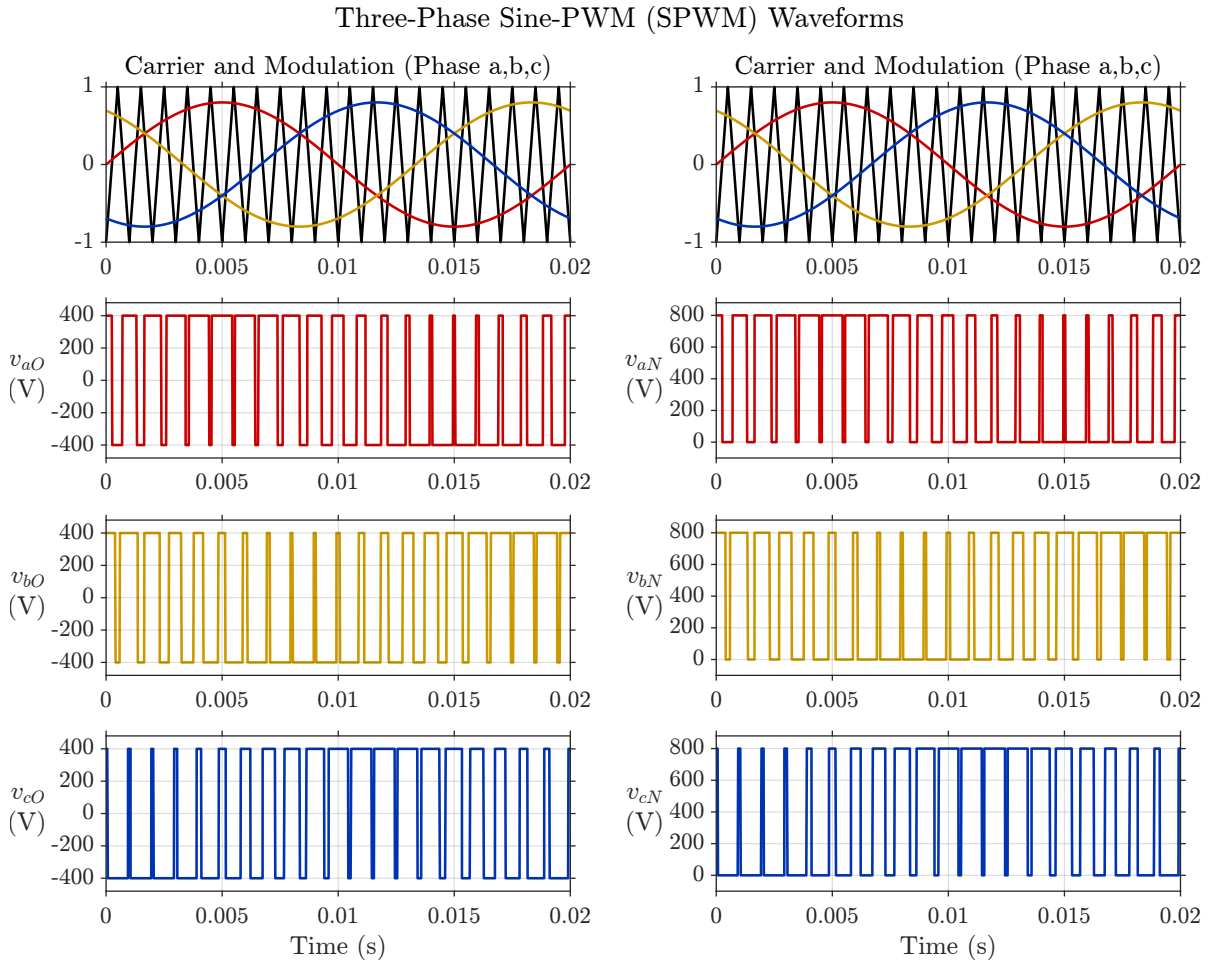


Figure 3: Three-phase Sine-PWM (SPWM) waveforms

Note: To determine whether statements (iii) and/or (iv) are correct, run an open-loop MATLAB/SIMULINK simulation of the three-phase inverter using Sine-PWM. Refer to the circuit diagram provided in Fig. 2.